

- 2 Red deer are herbivores common in the highlands of Scotland. They have no natural predators and in some areas culling of the deer is carried out in order to manage their numbers and maintain the species diversity of the ecosystem.

A group of scientists read a report which suggested that allowing sheep to graze alongside deer significantly reduces the need to cull the deer. They decided to carry out a survey to investigate the number of deer found grazing at 16 different sites.

- 8 sites contained deer **and** sheep.
- 8 sites contained deer but **no** sheep.
- Each site was visited during the first week of April and the first week of July in 2007 and again in 2009.
- Each site was 0.25 $\times$ 0.25 km.
- Within each site, three 10 $\times$ 10 m study plots were selected at random.
- The scientists used dung counts along a 1 $\times$ 10 m strip around each study plot to provide an estimate of the number of deer and sheep present. Dung is a pile of animal faeces.

Fig. 2.1 shows one study plot.

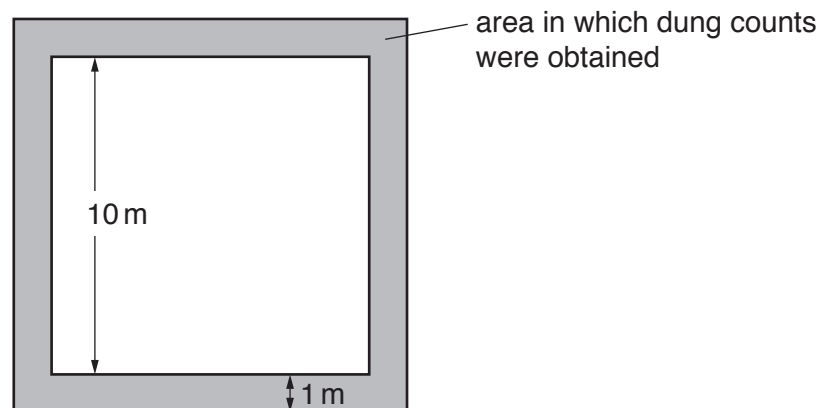


Fig. 2.1

(a) Describe a method that the scientists could have used to select the study plots randomly.

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..... [2]

(b) Suggest why counting the dung may **not** provide an accurate estimate of deer or sheep numbers.

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(c) State two variables that were standardised by the scientists.

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..... [2]

(d) The scientists also counted the number of plant species present within each of the 16 sites.

Fig. 2.2 shows the mean deer dung count for each site and the number of plant species present within each site.

Fig. 2.3 shows the mean deer dung count for the sites containing deer **and** sheep and for the sites containing deer but **no** sheep. Range bars are also included.

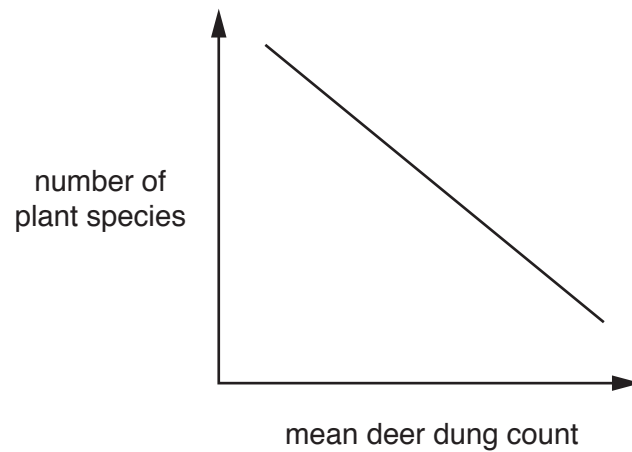


Fig. 2.2

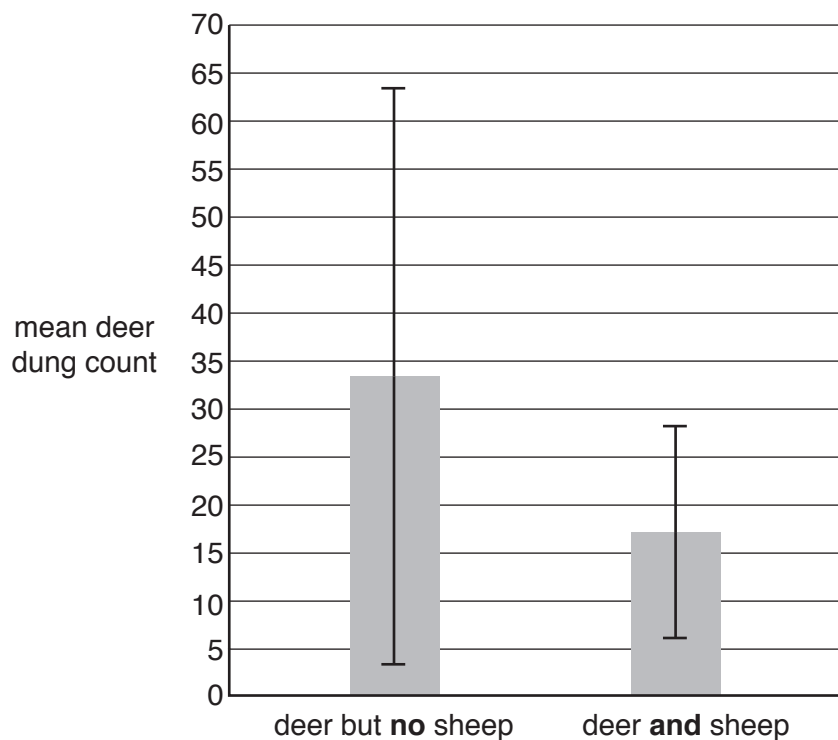


Fig. 2.3

The report that the scientists read suggested that allowing sheep to graze alongside the deer significantly reduces the need to cull the deer in order to maintain species diversity.

Explain why the results shown in Fig. 2.2 and Fig. 2.3 do **not** support this suggestion.

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[Total: 8]